

Crush Injury and Crush Syndrome

Guideline Title: Crush Injury and Crush Syndrome

Guideline Number: GUID-3203-T003

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 6/8/2021	Developed by: Air Care Team
Approved by: Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

XCVI. **PURPOSE:** A crush injury is a direct injury resulting from pressure. Crush syndrome is the manifestation of rhabdomyolysis caused by prolonged continuous pressure on muscle tissue. The goal is to identify crush syndrome and begin therapy early to prevent cardiac decompensation by early recognition and management of hyperkalemia

XCVII. **DEFINITIONS:**

XCVIII. **DEPARTMENT GUIDELINE:**

- A. Perform a general trauma assessment. Evaluate and support airway, breathing, and ventilation concerns.
- B. Assess for absent pulse or weak rapid pulse, capillary refill, pale, clammy, or cool skin, unusual pain, and signs of shock.
- C. Initiate and ensure adequate IV/IO access.
- D. Review or consider obtaining a 12-lead EKG.
- E. Keep patient warm.
- F. Consider infusing warm NS/LR wide open for 1L adult or 20ml/kg (pediatric) prior to or shortly after extrication. Use caution in setting of congestive heart failure and end-stage renal disease.
- G. If Creatinine Kinase levels greater than 6000 IU/L or a large mass of skeletal muscle (one or more limbs, with or without trunk or pelvis) has been crushed for **4 or more hours**, consider **Sodium Bicarbonate**
 - i. If prior to extrication
 1. Adult: **Sodium Bicarbonate** 8.4% (50 mEq) IV x 1.
 2. Pediatric: **Sodium Bicarbonate** 8.4% 1 mEq/kg IV x 1, max dose 50 mEq.
 3. Infant: **Sodium Bicarbonate** 4.2% 1 mEq/kg x 1.
 - ii. If following extrication
 1. Adult: **Sodium Bicarbonate gtt.** Mix 40mEq of 8.4%/250ml D5W (remove 40ml of fluid from bag prior to adding bicarb) and administer infusion 250ml/h.
 2. Pediatric: Mix 40mEq of 8.4%/250ml D5W (remove 40ml of fluid from bag prior to adding bicarb) and administer infusion at maintenance fluid rate per *GUID-PED001 – General Management of the Pediatric Patient*.
- H. Immobilize the limb. Do not elevate.

- I. If signs of inadequate perfusion, treat per *GUID-M011 - Hypotension*.
- J. If hyperkalemia is known or suspected (bradycardia, QRS > 0.12, or peaked T-waves), treat per *GUID-M006 - Electrolyte Disorders – Potassium Imbalance*.
- K. Treat pain per *GUID-G005 - Analgesia and Sedation Management* and anxiety per *GUID-G006 Anxiety- Combative Patient Management*.
- L. If after discussion with those providers leading the extrication attempt it is determined that the time required for extrication will be a risk to the patient's life, the flight team should consider contacting medical control regarding bringing an appropriately trained physician to the scene to consider amputation of an entrapped limb.

XCIX. **DOCUMENTATION:**

- A. EMS Charts

C. **REFERENCES:**